



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP2613 - 01

SYSTEM ANALYSIS AND DESIGN

ASSIGNMENT 1

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ITEM 1 - PROBLEM DEFINITION

a) Current Process:

MediCare Plus Clinic's appointment booking system currently uses traditional and inefficient ways such as phone calls, WhatsApp messages, paper appointments and Excel files.

b) Current Problem:

1. Duplicate Booking
2. Long waiting Times
3. Difficulty tracking patient queue status
4. Delayed reporting
5. Complication to access booking information

c) Formal Problem Statement:

The problem regarding this system is that when the clinic's patient count is increasing, many issues will rise. For example, the clinic will face duplicate booking. The reasons are this clinic consists of various booking platforms which will cause duplication of booking. This will also cause longer waiting times due to the clinic needing to settle problems that happen real-time. Similarly, having too much booking platform causes the clinic to have complications to track patient queue and problems to access booking information and because of all the problems stated before, the clinic's reporting will be delayed.

ITEM 2 - OBJECTIVES, REQUIREMENTS AND CONSTRAINTS

a) Objectives

1. To avoid duplicate bookings by using real time synchronization slot availability
2. To provide live tracking for queue status updates
3. To reduce patient waiting time by at least 25%
4. To generate report automatically without consuming longer time

b) Functional requirements

1. The system should allow patients to create a profile and register.
2. The system should enable patients to book, reschedule or cancel the appointment through the online platform.
3. The system should prevent double booking by using real time slot availability.
4. The system should automatically assign a queue number after patients check-in.
5. The system should have a display of current queue status.
6. The system should send a reminder to the patient the day before the appointment automatically.

c) Non-functional requirements

1. The system can be accessed by staff and patients at all times.
2. The system should give a fast response, within 5 seconds.
3. The system should protect the patients data by using data encryption.
4. The system should have accurate live-tracking updates.

d) Constraints

1. The system must be developed using allocated hardware budget of RM15000 and software of budget RM20000.
2. The system must be completed within the allocated project timeframe.
3. The system must be running without disrupting the clinic daily operations
4. The system must be developed by using technologies that can be maintained by existing staff.

ITEM 3 - FEASIBILITY STUDY

a) Technical feasibility

The system will use a web-based technology. The clinic can have its own cloud as hosting services because it is more affordable rather than to buy servers. This technology is easily accessible and well-documented. The development of the system can be done by using open-source software like MySQL to minimize software licensing costs. Desktop computers that already exist in the clinic can be used to access the system through standard web browsers. Therefore, this project is technically feasible.

b) Operational feasibility

The new system is designed to be user-friendly. The current system manual methods such as entering the appointment details manually which required more work can be time-consuming for the staff. The proposed solution will help reduce the workload and make the appointment booking much easier. Clinic staff must be able to use the system after a short training period. Patients will benefit from the online booking and real-time queue visibility. The system will not disrupt the clinic's daily operations. Therefore, this project is operationally feasible.

c) Economic feasibility

Based on the Cost-Benefit Analysis (CBA), the profitability index (PI) is 0.16, which is less than 1.0. The development cost is RM 38,500, and the total gain or loss after three years is a negative RM 5,998.36. However, the project also offers intangible benefits that are not reflected in the financial calculation, such as improved patient satisfaction, reduced staff workload, more accurate reporting, and better decision-making for clinic management. Therefore, the project is partially feasible.

d) Overall feasibility decision

After looking at all three feasibility studies, this project is partially feasible.

Technically, the project is feasible because the technology needed is easy to get and affordable. The clinic can use cloud hosting and open source software. The existing computers at the clinic are good enough to run the system.

Operationally, the project is feasible because staff can be trained in two days. Patients will find the system easy to use with online booking and queue visibility. The clinic management also supports this project.

Economically, the project is not good because the Profitability Index is only 0.16, which is far below 1.0. The development cost is RM 38,500 and the financial return is negative.

Overall, the clinic should not proceed with this project unless they can reduce the cost significantly or find more benefits. Without major changes, the project is not worth the investment.

ITEM 4 - ECONOMIC FEASIBILITY : COST - BENEFIT ANALYSIS (CBA)

CRITERIA	Year 0	Year 1	Year 2	Year 3
1. COST				
A. Development				
i. Hardware	16500	-	-	-
ii. Software	22000	-	-	-
Total	38500	-	-	-
B. Operation				
i. Maintenance	-	5500	5775	6063.75
ii. Support & Upgrade	-	11000	11550	12127.50
Total	-	16500	17325	18191.25
Present Value (PV)	-	15277.78	14853.40	14440.80
Accumulated Cost	-	53777.78	68631.18	83071.98
2. Benefits				
i. Interest Investment	-	10800	11232	11681.28
ii. Consultation Services	-	18000	18720	19468.80
Total	-	28800	29952	31150.08
Present Value (PV)	-	26666.67	25679.01	24727.94
Accumulated Benefits	-	26666.67	52345.68	77073.62
Gain or Loss	-	(27111.11)	(16285.50)	(5998.36)
Profitability Index	0.16			

$$\text{Profitability Index} = \frac{5998.36}{38500}$$

$$= 0.16$$

Profitability Index = 0.16, showing that it is not a good investment because its index is less than one.

ITEM 5 - CONCLUSION

In conclusion, MediCare Plus Clinic currently faces inefficiencies and difficulties in managing appointments due to manual processes, resulting in various problems such as duplicate bookings, long waiting times, and difficulty in tracking patient queues. The proposed Clinic Appointment and Patient Queue Management System aims to improve efficiency by automating appointment scheduling through cloud systems, MySQL and web-browsing, providing real-time queue tracking, and enhancing data accessibility to ensure that the system is more efficient and simpler to use.

Based on the feasibility study, the system is technically and operationally feasible as the required technology is available and users can adapt to the system with minimal training up to a total of 16 hours in the span of 2 days. However, the economic feasibility analysis calculated from the Cost-Benefit Analysis (CBA) shows that the project is not financially viable, with a profitability index (PI) of 0.16, which is less than 1. Therefore, the project is considered partially feasible. The proposed project should not proceed unless the economical feasibility improves. Hence, it is recommended that the clinic reconsider the cost structure and explore more cost-effective solutions before finally proceeding with full implementation.